Session 1:
NLP Foundations

SMM694 Applied Natural Language Processing

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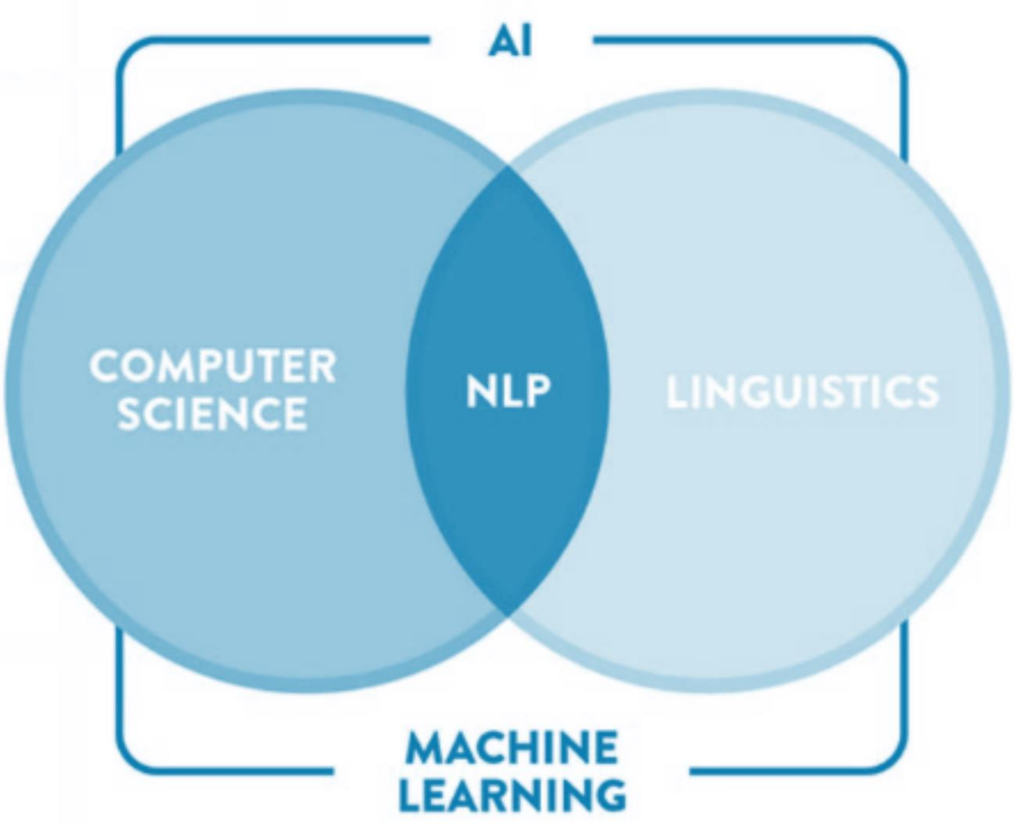
What is Natural Language Processing?

- **Definition:** Field of Artificial Intelligence focusing on interaction between computers and humans through natural language.
- NLP intersects **Computer Science**, **Linguistics**, and **Machine Learning**.



The Purpose of NLP

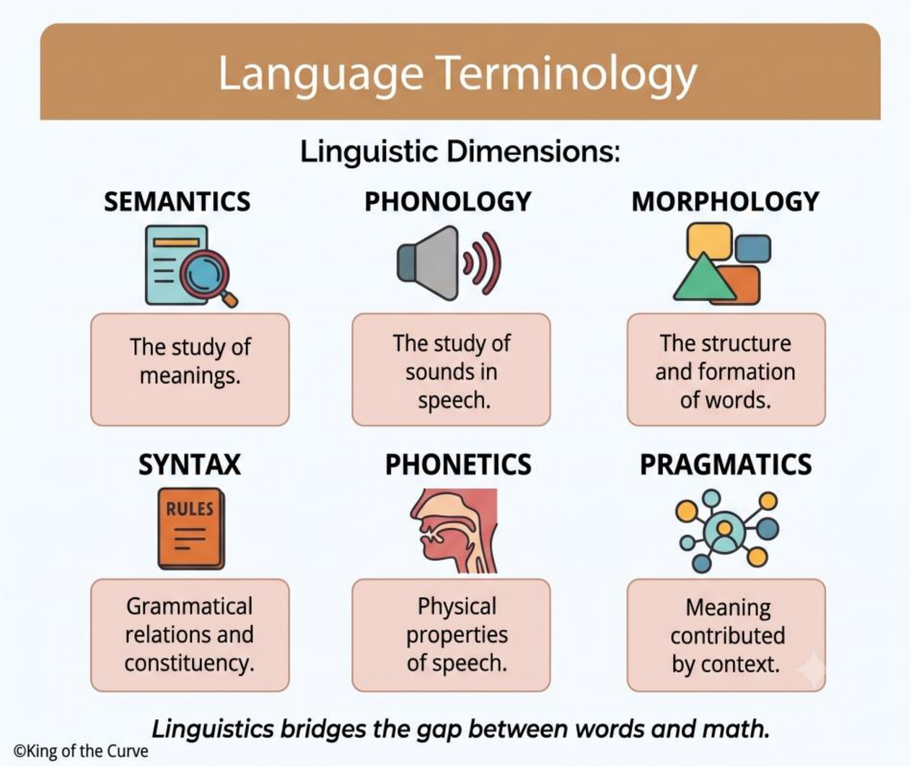
- **Primary Goal:** Enable computers to understand, interpret, and generate human language.
- Output must be both **meaningful** and **useful** for real-world applications.

Conceptual Foundations

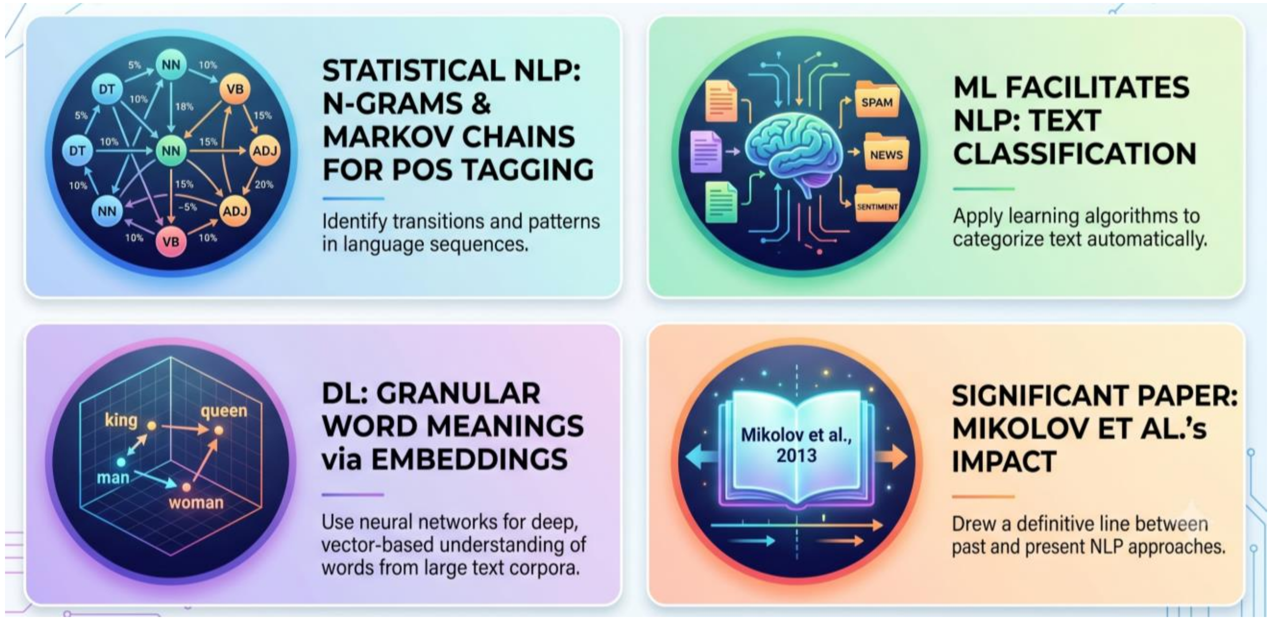
Linguistic Dimensions:

- **Pragmatics:** Meaning contributed by context.
- **Phonetics:** Physical properties of speech.
- **Semantics:** The study of meanings.
- **Syntax:** Grammatical relations and constituency.

Linguistics bridges the gap between words and math.



Technical Foundations



- **Statistical Analysis:** n-grams, Markov Chains for POS tagging.
- **Machine Learning:** Facilitates classification tasks.
- **Deep Learning:** Granular word representation based on large corpora.
- **Milestone:** Mikolov et al.'s paper marks the shift to modern AI-driven NLP.

Why NLP Matters

Conversational AI

Powering tools like ChatGPT for fluid human-like interaction.

Machine Translation

Google Translate and DeepL breaking linguistic barriers.

Sentiment Analysis

Monitoring social media and customer feedback in real-time.

Speech Recognition

Enabling Siri, Alexa, and Google Assistant voice commands.

Text Summarization

Automating long-form content distillation for analysts.

Question Answering

Chatbots providing instant, accurate support solutions.

Impact: Enhanced communication and task automation.

Development: Early Roots

1949
Weaver's Memorandum

1960s
Grammar Theories

1970s
Conceptual Ontologies

1980s
Symbolic Models

1990s
Statistical Models

Rule-based systems dominated until data-driven statistical methods gained prominence.

Modern Development Timeline

2003
Neural Language Models

2008
Multi-task Learning

2013
Word Embeddings

2015
Attention & Seq2Seq

2018
Pretrained (BERT/GPT)

Transition to deep neural networks and attention mechanisms redefined the field.

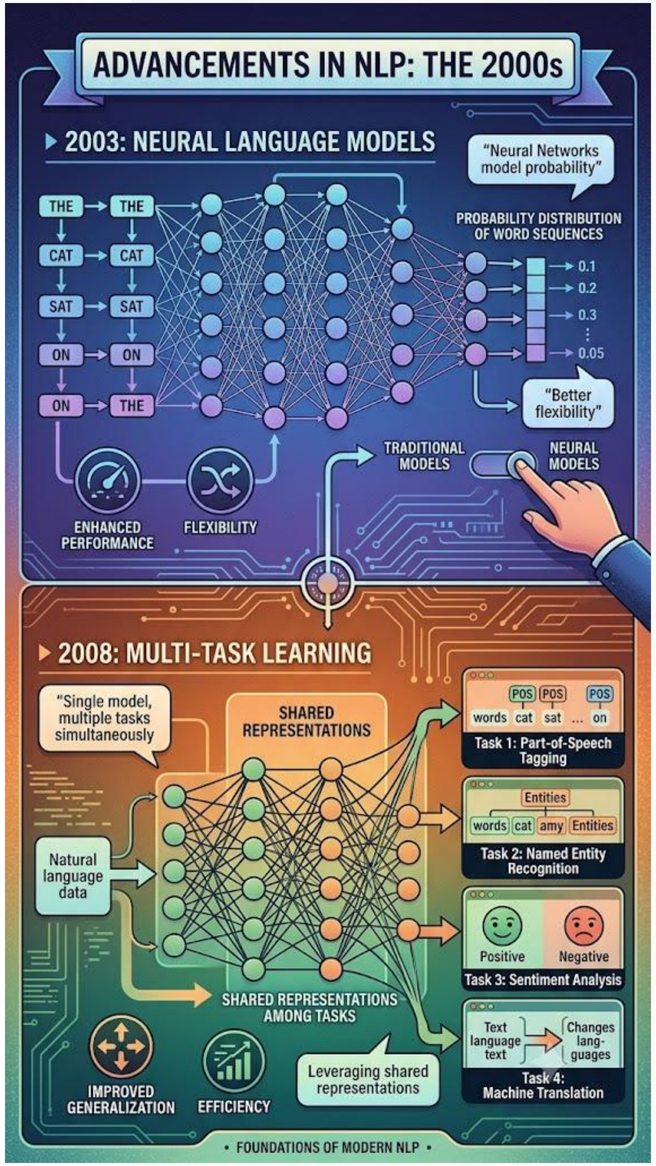
2000s: A Paradigm Shift

2003: Neural Language Models

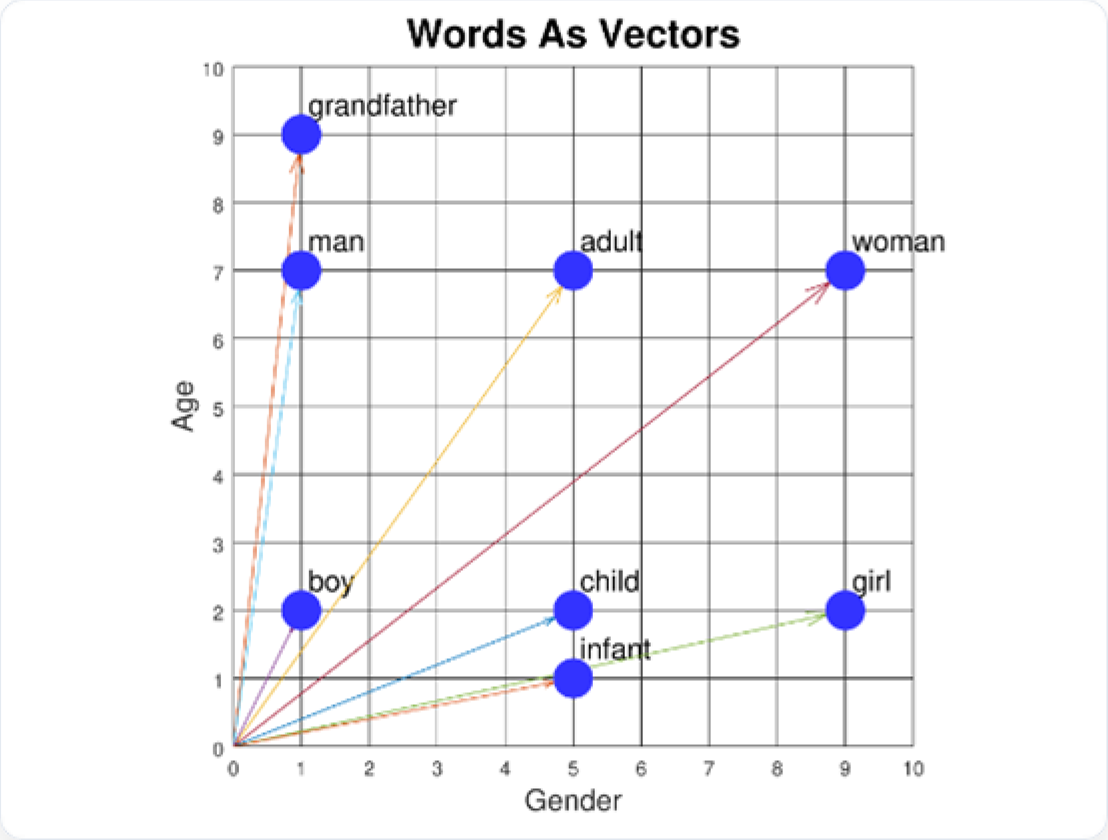
Neural networks used to model word sequence probability distributions, offering superior flexibility.

2008: Multi-task Learning

Training models on multiple tasks simultaneously to improve generalization and efficiency.

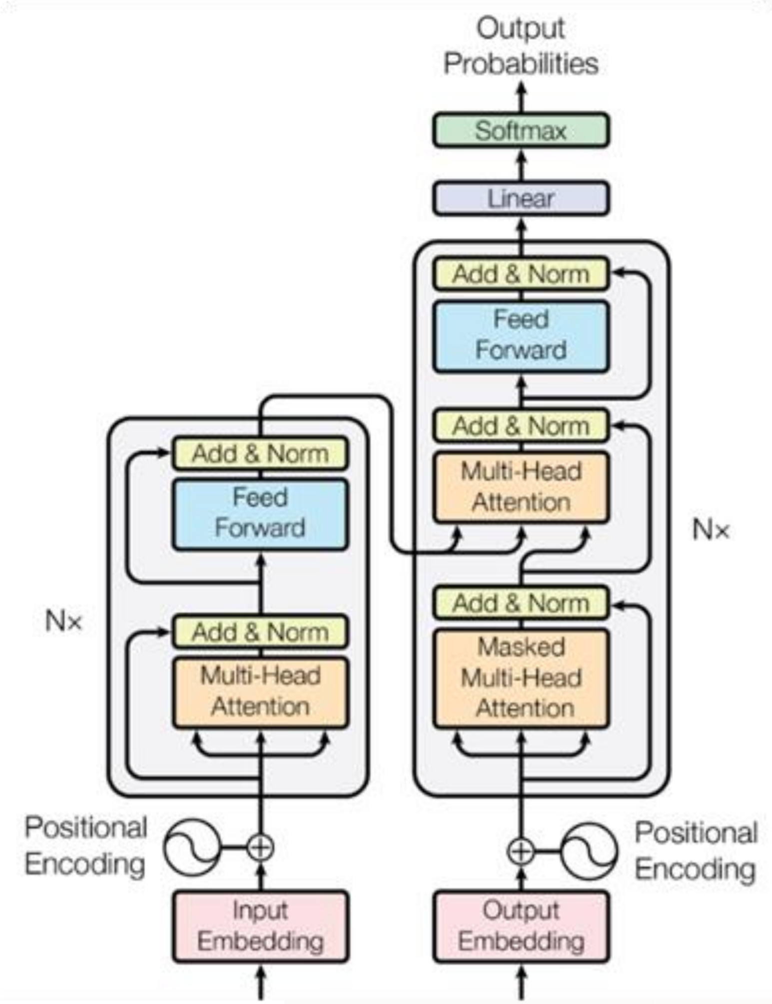


2013-2014: Representation



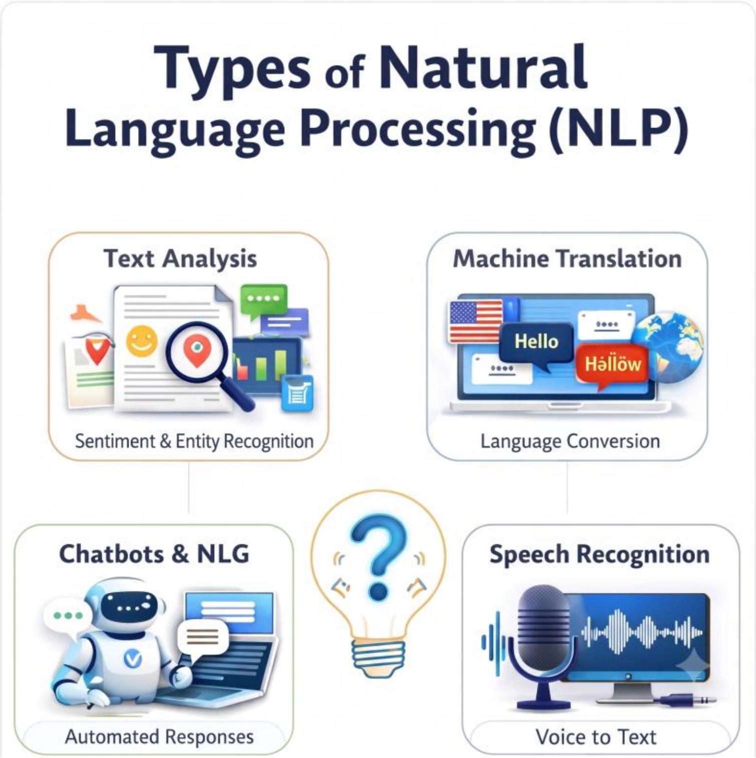
- **2013 Word Embeddings:** Word2Vec provided dense vectors capturing semantic relationships.
- **2013 NLP Neural Nets:** Prevalence of CNNs and RNNs for sequential data.
- **2014 Seq-to-Seq:** Enabled generation from sequences (translation/summarization).

2015–2018: Scale & Transformers



- **2015 Attention:** Selective focus on input parts; groundwork for Transformer.
- **2018 Pretrained Models:** BERT and GPT dominate benchmarks.
- Models are pre-trained on massive datasets and fine-tuned for specialized tasks.

Operational Applications



- **Text Summarization:** Distilling long documents.
- **Speech Recognition:** Transforming voice to text.
- **Dialogue Systems:** Customer relationship CRM.
- **Text Translation:** Multilingual capabilities.
- **Phonetics:** Application of speech sounds.

Strategic Applications

Vector Semantics

Understanding vocabulary in complex context.

Topic Modeling

Revealing latent themes across text corpora.

Sentiment Analysis

Summarizing attitudes in social media/reviews.

Text Categorization

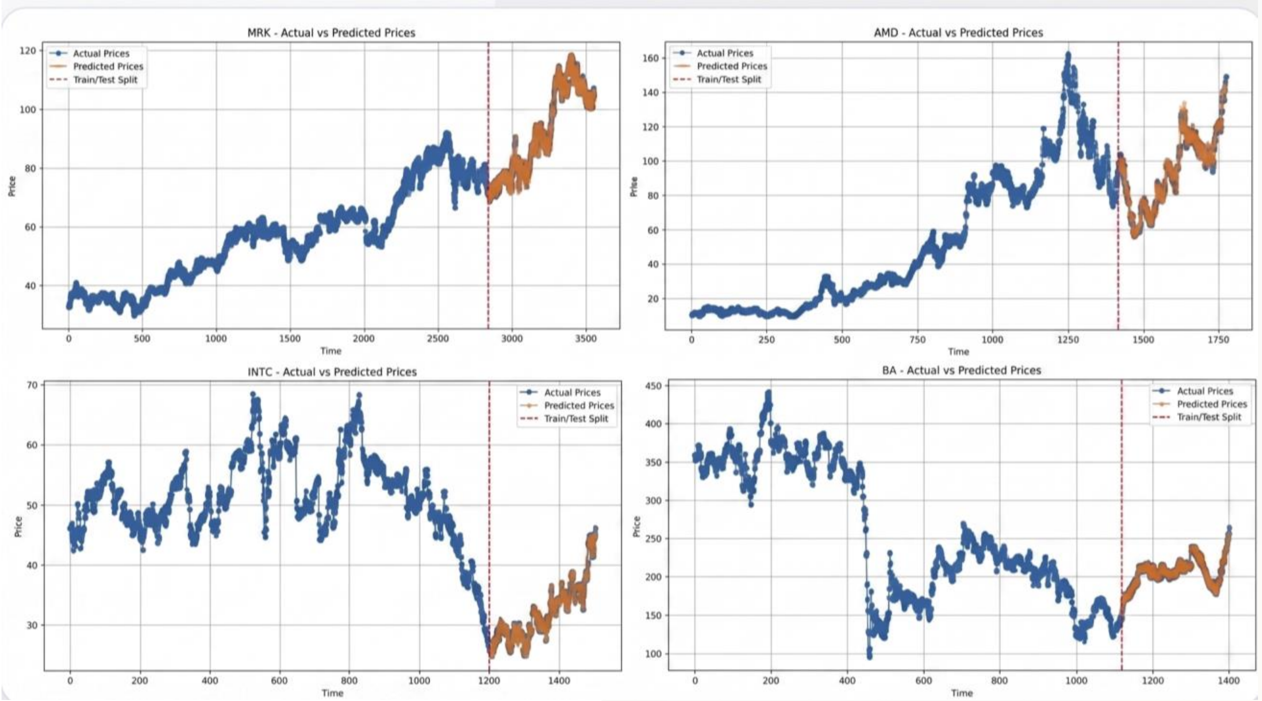
Automated grouping for granular analysis.

Information Extraction

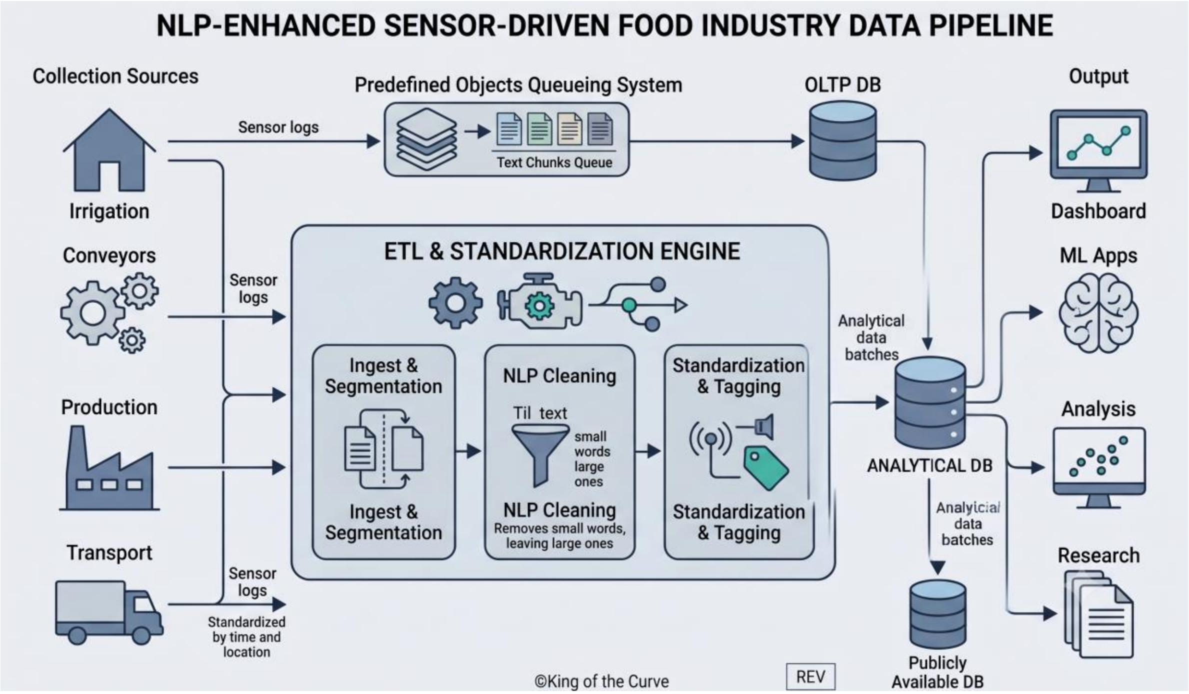
Tracing entities, events, dates, and relationships across vast datasets.

Applications in Finance

- Market Sentiment Analysis
- Stock Price Prediction
- Fraud Detection
- Customer Feedback Analysis
- News Sentiment Impact



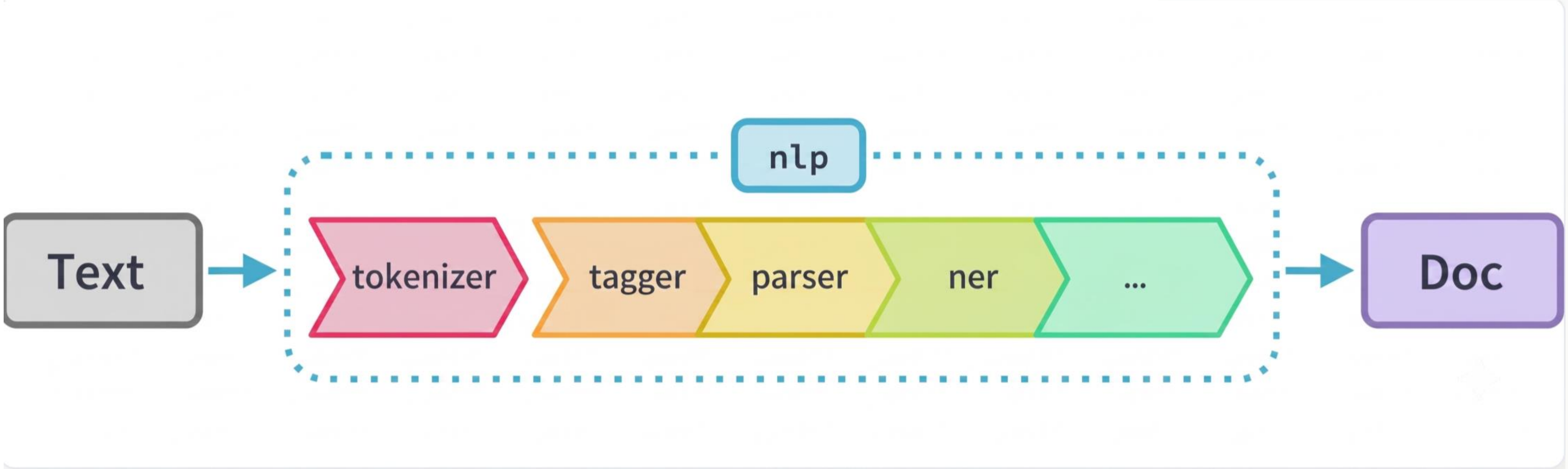
The NLP Pipeline



Benefits of Pipelines:

- **Standardization:** Tags words with metadata.
- **Dimensionality Reduction:** Lemma-based replacements.
- **Manageable Input:** Segments long documents.
- **Cleaner Data:** Precise stop-word removal.

Pipeline Components



Tokenization

Splitting text into segments.

Tagging

POS association.

Parsing

Syntactic dependencies.

Tokenization Example

"Apple is looking at buying U.K. startup for $1 billion"

```
In [29]: doc4 = nlp(u"Apple isn't looking at buying U.K. startup.")

         for token in doc4:
             print(token.text, token.pos_)

Apple PROPN
is VERB
n't ADV
looking VERB
at ADP
buying VERB
U.K. PROPN
startup NOUN
. PUNCT
```

Text is split into meaningful units (tokens) for downstream processing.

Pipeline Step: Tagging

- Assigns part-of-speech (POS) tags to tokens, identifying their grammatical roles.
- TAG (Detailed Part-of-Speech Tag) provides a more detailed part-of-speech tag.
- Example:

TEXT	LEMMA	POS	TAG	DEP	SHAPE	ALPHA	STOP
Apple	apple	PROPN	NNP	nsubj	Xxxxx	True	False
is	be	AUX	VBZ	aux	xx	True	True
looking	look	VERB	VBG	ROOT	xxxx	True	False
at	at	ADP	IN	prep	xx	True	True
buying	buy	VERB	VBG	pcomp	xxxx	True	False
U.K.	u.k.	PROPN	NNP	compound	X.X.	False	False
startup	startup	NOUN	NN	dobj	xxxx	True	False
for	for	ADP	IN	prep	xxx	True	True
$	$	SYM	$	quantmod	$	False	False
1	1	NUM	CD	compound	d	False	False
billion	billion	NUM	CD	pobj	xxxx	True	False

Pipeline Step: Parsing

- Annotates tokens with syntactic dependency labels, defining relationships between words.
- Example

TEXT	LEMMA	POS	TAG	DEP	SHAPE	ALPHA	STOP
Apple	apple	PROPN	NNP	nsubj	Xxxxx	True	False
is	be	AUX	VBZ	aux	xx	True	True
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$	$	SYM	$	quantmod	$	False	False
1	1	NUM	CD	compound	d	False	False
billion	billion	NUM	CD	pobj	xxxx	True	False

Named Entity Recognition (NER)

- Identifies and classifies named entities in text, such as people, organizations, and locations.
- Example:

Token	POS	DEP	Entity Type
Apple	PROPN	nsubj	Organization
is	AUX	aux	none
looking	VERB	ROOT	none
at	ADP	prep	none
buying	VERB	pcomp	none
U.K.	PROPN	compound	Geopolitical Entity
startup	NOUN	dobj	none
for	ADP	prep	none
$	SYM	quantmod	Monetary Value
1	NUM	compound	Monetary Value
billion	NUM	pobj	Monetary Value

Python Ecosystem for NLP

NLTK

Academic swiss-army knife; low memory.

spaCy

Industrial strength, efficient and reliable.

Gensim

Specializes in embeddings and topic models.

TextBlob

High-level sentiment and translation.

Transformers

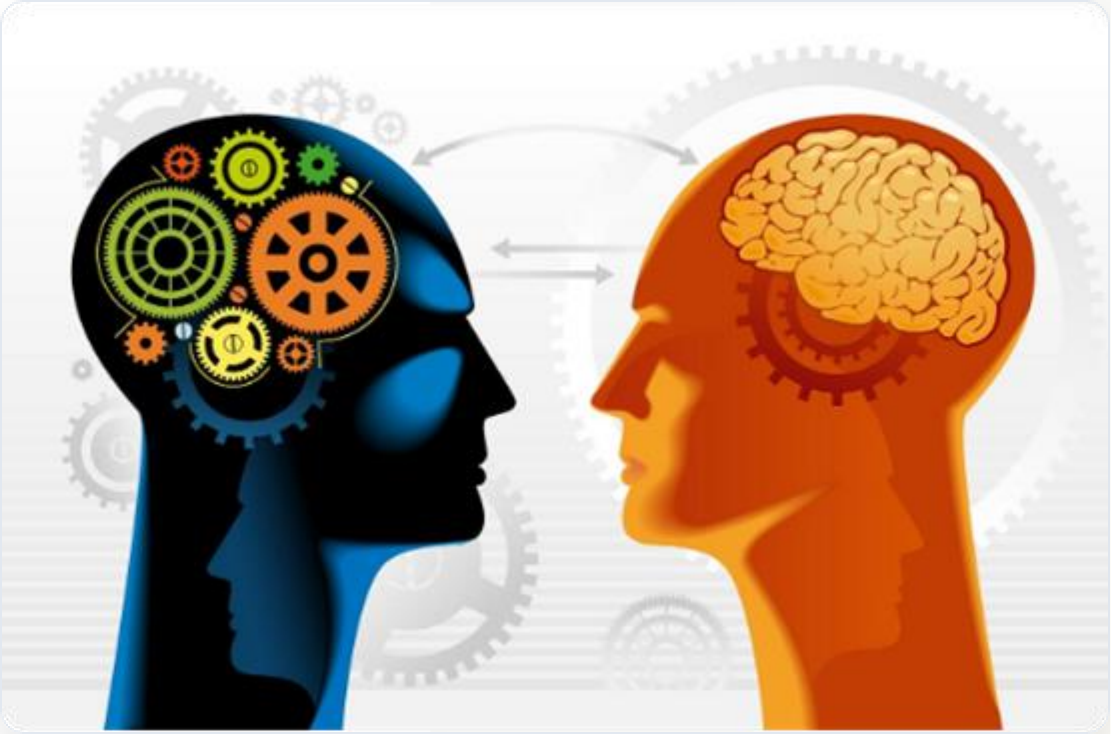
HuggingFace's SOTA BERT/GPT framework.

scikit-learn

General ML tools for text vectors.

Core Challenges in NLP

- **Ambiguity:** Multiple meanings of words.
- **Context:** Shifting significance across phrases.
- **Data Sparsity:** Lack of labeled datasets.
- **Domain Specificity:** Jargon and terminology.
- **Culture:** Dialects, slang, and biases.



Future Horizons

Multimodal NLP

Integrating text with audio and images.

Ethical AI

Addressing bias and ensuring transparency.

Low-Resource

Inclusivity for less-documented languages.

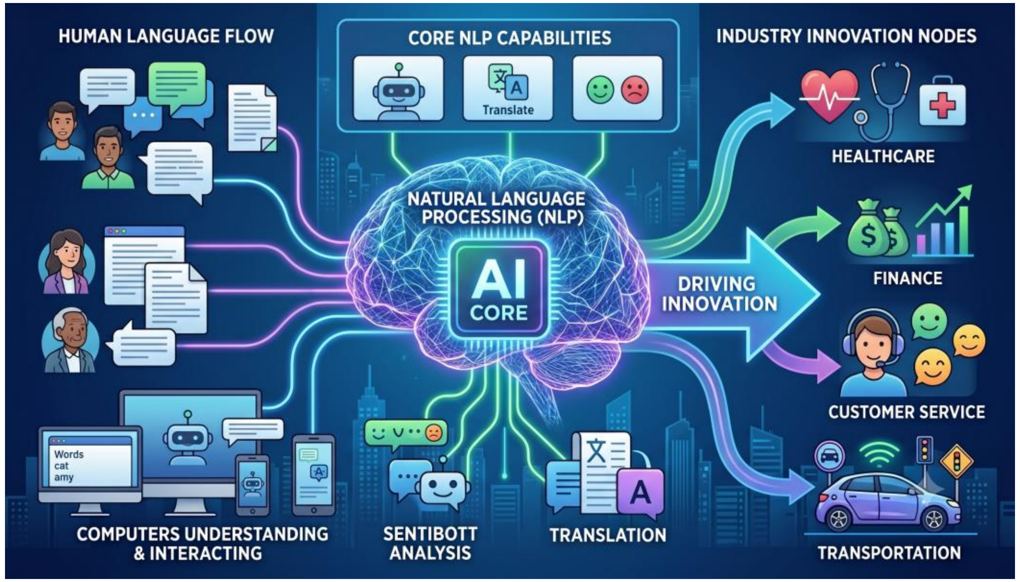
Explainability

Understanding model decision paths.

Efficient Learning

Continued progress in Transformers for smaller, faster generation.

Concluding Remarks



Recap: We've explored NLP fundamentals, pipelines, historical roots, and current applications.

Call to Action: Stay curious. NLP is a rapidly evolving field driving innovation across every industry.

Questions?

Thank you for your attention.

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